

EDUCATION

Brown University

2024–2028 Providence, United States

Concentrating in Engineering Physics.

Courses Taken: Deep Learning for Scientists & Engineers, Engineering Statics and Dynamics, Data Engineering, Applied Ordinary Differential Equations, Multivariable Calculus (Physics/Engineering)

WORK EXPERIENCE

CERN Machine Learning Research Intern

June–August, 2025 Geneva, Switzerland

- Developed and proposed a calibration technique for Higgs boson decay simulations in electron–positron collisions using Graph Neural Networks and Transformers (PyTorch; 30m+ parameters).
- Collaborated with Brown, MIT, and CERN professors to characterize forward–backward asymmetry in FCC-ee, informing detector design optimizations with 1mrad precision.

Kyron Medical Machine Learning Research Intern

May–August, 2025 Geneva, Switzerland

- Led the design and development of an ML-powered pipeline leveraging large language models (LLMs) to automate and streamline denied claim appeals, trained on over \$4.5M in denied claims data.

Ramallah Municipality Systems Engineer Intern

May–August, 2024 Ramallah, Palestine

- Designed and deployed automated check-in workflows and server maintenance pipelines, cutting downtime and boosting operational efficiency by 40%.
- Delivered full-stack systems support (infrastructure + helpdesk), ensuring reliable municipal IT operations across multiple departments.

Global Tech Systems Engineer Intern

May–August, 2023 Ramallah, Palestine

- Architected and migrated core server infrastructure to a scalable, cost-optimized environment, reducing infrastructure expenses by 30%.

RESEARCH/PERSONAL PROJECTS

Predictive Modeling of Charged Particle Dynamics using Physics-Based Neural Networks 2024–2025

- Compared and applied Symplectic Networks (SympNets), Physics Neural Networks (PNNs), and Physics-Informed Neural Networks (PINNs) to model charged particle trajectories in electromagnetic fields with high accuracy.

AI & Robotics Development

2017–2025

- Built a self-driving car integrating Python-based AI with Arduino hardware to demonstrate autonomous navigation.
- Implemented and released an open-source AI modeling library in C, adopted by 50+ developers worldwide.
- Designed and deployed a drone system for a 15,772 square meter solar panel field, currently in use at King’s Academy in Jordan.

LEADERSHIP EXPERIENCE

WorkFrame Consulting Firm CEO

2022–2024 Palestine

- Founded and led a consulting firm offering financial, construction, and international strategy support to small businesses, achieving a 45% cost reduction through importation and operational advice.
- Advised a farm startup in scaling operations to a monthly output of 100 sheep through targeted business consulting.

Linguistic Founder

2022–2024 Jordan

- Developed gamified educational platforms—including websites and mobile apps—to enhance language learning engagement and retention, piloted by 4+ schools in Jordan.
- Presented the startup in pitch competitions and secured partnerships with educational institutions across Jordan.

Mushrooming Foundation Founder

2022–2024 Jordan

- Founded a nonprofit initiative to expand access to programming education across Jordan.
- Led the organization of national hackathons and authored programming curricula adopted by 10+ partnering schools.

TEDx Talk Speaker

March–May 2024 Amman, Jordan

- Delivered a TEDx talk with 500 attendees and 1000+ online views titled “Beyond the Weld: Unraveling Society’s Hidden Fusions,” exploring how industrialization and automation cause society to undervalue human effort.

SKILLS

Languages: Arabic (Native), English (Native), Korean (Intermediate), Italian (Intermediate)

AI/ML: PyTorch, SympNet, PINN, Reinforcement Learning, DDQNs, Machine Vision, LLMs

AWARDS & HONORS

Robotics Engineering Excellence (2024), First Place – Samsung Innovation Coding and Icode Edu (2024)